



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Preservation Infrastructure and the Problem of Digital Heritage Loss

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Abstract

This research proposes "digital taphonomy" as a methodological framework to address a critical gap in 3D cultural heritage preservation. While the current FAIR-compliant infrastructure development and research in the field focuses on documenting successful acquisitions, the systematic approaches for learning from preservation failures remain underdeveloped. The emphasis of digital taphonomy on heritage data is purely technical storage, but this study attempts to integrate governance and semantic dimensions that are often overlooked in current practice.

Scholars have documented some challenges in the current approach. These include technological obsolescence that renders 3D models uninterpretable (Paquet et al., 2005; Amico & Felicetti, 2024), and the conflict of ethical obligations with legal-practical constraints (Clarke, 2015). Recent literature emphasises that long-term usability requires more than bit-level preservation, necessitating the documentation of uncertainty and the use of "paradata" to distinguish between original data and operator interventions (Collina et al., 2025; Kruse, 2024). Furthermore, the shift toward "Memory Twins" highlights the importance of layering metadata and intangible values to prevent semantic loss (Cassar et al., 2025).

A systematic review of literature (2010–2026) is used in this study to develop a four-category failure typology based on OAIS foundations: (a) Technical (software dependency), (b) Institutional (organisational continuity), (c) Semantic (interpretative context, e.g., Bajena, 2025), and (d) Governance (community representation and access). This research transforms undocumented data loss into systematic learning opportunities, delivering a preliminary digital taphonomy framework, a failure typology with application guidelines, and recommendations for integrating failure analysis into FAIR-oriented workflows. By repositioning failure as actionable knowledge, this work supports the development of sustainable, community-inclusive infrastructure for 3D heritage, directly contributing to the FAIR 3D Heritage conference session on web-based publication infrastructures and preservation strategies.

References

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